

Booleans and Comparisons

Truth values, comparisons, and explicit conversions

Gregory S. DeLozier, PhD

Breaking Changes

New keywords: `true`, `false`, `boolean`, `type`

Rename earlier variables that use those exact names.

`true_value`, `boolean_value`, and `typewriter` remain identifiers.

Computing an Answer

```
__input = "21";  
answer = number(input()) * 2;  
large_enough = answer >= 40;  
print(large_enough);
```

```
true
```

A New Literal Node

```
ready = true;
```

```
{"tag": "boolean", "value": True}
```

Vertex spelling: true. Python value: True.

Comparing Values

Expression	Result
<code>42 == 42.0</code>	true
<code>"dog" != "cat"</code>	true
<code>-1.5 < .5</code>	true
<code>2 <= 2</code>	true
<code>2 > 3</code>	false
<code>3 >= 3</code>	true

Ordering Strings

Expression	Result
"ant" < "bee"	true
"ant" < "anteater"	true
"ant" <= "ant"	true
"Z" < "a"	true
"10" < "2"	true

Compare characters, not numeric meanings.

Types Still Matter

Expression	Result
<code>true == 1</code>	false
<code>"1" == 1</code>	false
<code>"1" < 2</code>	Type error
<code>true + 1</code>	Type error
<code>number(true) + 2</code>	3

Explicit Conversions

Expression	Value
<code>number(true)</code>	<code>1</code>
<code>string(false)</code>	<code>"false"</code>
<code>boolean(-2)</code>	<code>true</code>
<code>boolean(0)</code>	<code>false</code>
<code>boolean(" FALSE ")</code>	<code>false</code>

Text Needs an Interpretation

<code>boolean("false")</code>	<code>false</code>
<code>boolean("1")</code>	<code>value error</code>
<code>boolean(number("1"))</code>	<code>true</code>

No nonempty-string truthiness.

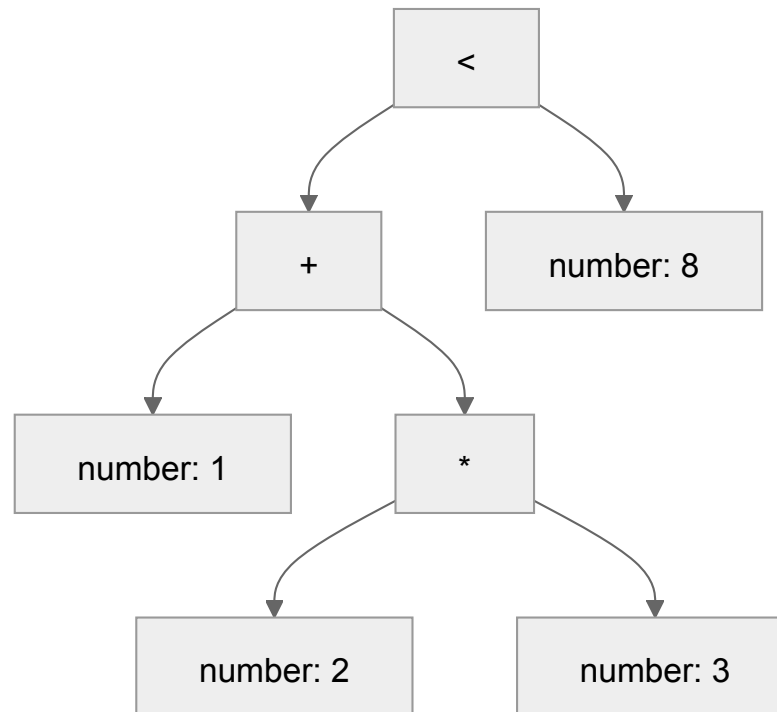
The Type of a Value

Expression	Result
<code>type(42)</code>	<code>"number"</code>
<code>type(.5)</code>	<code>"number"</code>
<code>type("42")</code>	<code>"string"</code>
<code>type(1 < 2)</code>	<code>"boolean"</code>

`type(input())` still reads input.

The Comparison Tree

1 + 2 * 3 < 8



One Comparison at a Time

```
expression ::= comparison
comparison ::= arithmetic_expression
              [ compare_op arithmetic_expression ]
```

Expression	Result
1 < 2 < 3	Syntax error
(1 < 2) == true	true
(1 < 2) < 3	Type error

Checking the Implementation

```
bash run-tests.sh
```

All six comparisons. Exact result types. Conversion boundaries.

Input arguments run once; comparison operands run left to right.

Practice

```
true == 1  
number(true) == 1  
boolean("0")  
boolean(number("0"))
```

Which succeed, and what does each produce?